



Review

A hitchhiker's guide to the *Arabidopsis* ferromeWolfgang Schmidt^{a,*}, Thomas J. Buckhout^b^a Academia Sinica, Institute of Plant and Microbial Biology, Taipei, Taiwan^b Applied Botany, Humboldt University Berlin, Invalidenstraße 42, 10115 Berlin, Germany

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ABSTRACT

Unraveling the mysteries behind the perception and response to iron deficiency has resulted in a vast collection of data that is summarized under the topic “ferromics”. The analysis of the immediate effects induced by iron deficiency has been facilitated and in most cases greatly accelerated by the development of analytical and computational tools. These tools permit on the one hand the collection of information from a large number of sources and on the other the analysis of this collection to detect patterns in the re-ordering homeostatic processes at the genomic, transcriptomic, and proteomic levels. Deciphering the encrypted information from high-throughput datasets have become a major challenge in plant biology, but this information also sets the stage for a more complete, integrative view on how plants respond to a varying supply of iron.

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“The answer to the great question... of life, the universe and everything... is... forty-two,” said Deep Thought, with infinite majesty and calm.”

—Douglas Adams (*The Hitchhiker's Guide to the Galaxy*)

1. Introduction

Being the fourth most abundant metal in the Earth's crust, iron (Fe) is present in most environments in high concentrations, exceeding by far the requirements of the cell. Surprisingly, in spite of its high abundance in the environment, Fe deficiency is a significant medical and agricultural problem. Approximately two billion people suffer from Fe deficiency-induced anemia, which causes early termination of pregnancy, birth defects, retarded motor development and a general reduction in fitness. An analogous situation is found in plants. Fe deficiency is a major constraint to plant growth, affects the quality of edible plant parts, and is the most common nutritional disorder worldwide. The poor solubility of ferric oxyhydroxides, the prevailing form of Fe in aerobic soils, makes Fe biologically unavailable. Understanding how cellular Fe homeostasis is maintained would provide a basis for engineering plants with enhanced Fe concentration and would help to counteract malnutrition.

During the past decade, forward and reverse genetic approaches have greatly contributed to our understanding of the molecular mechanisms acclimating plants to the low availability of Fe. Individual components involved in the translation of the cellular Fe status into the expression of genes encoding the machinery of signaling, acquisition and distribution of Fe have been identified [1–4]. Putting any one of these components into its appropriate context is a challenging undertaking and can only be achieved by looking at many, if not all components in parallel and, preferentially, in a manner that permits the observation of dynamic interactions between them. Generally, systems biology is understood as a systematic approach for “incorporating information on all genes and proteins in a cell into a composite model of interacting components” [5]. A systems biology approach facilitates accounting for and integrating different levels of control, such as genetic and epigenetic regulation, post-transcriptional and post-translational changes and understanding the behavior of the system by collecting information about the dynamics of the interactions between individual components. Here, we review how comprehensive analyses using functional genomics technologies have allowed new insights into the mechanisms of cellular Fe homeostasis and how potentially hidden parts of the iceberg may be important for a full mechanistic understanding of the Fe deficiency response. We will also highlight underexplored areas that may hide a wealth of undiscovered facts on how Fe concentrations are sensed and translated into adaptive processes aimed at improving cellular Fe homeostasis. We do not, however, attempt to cover all aspect of

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Table 1

Complete list of the genes responding to Fe deficiency and found in the ferrome. The ATG number of genes presumably regulated by the FIT as described by Colangelo and Gueriot (Table 1 in [25]) are typed in bold and those determined by Long et al. (Table S3c in [27]) to be regulated by PYE are underlined.

ATG	Description
At1g01570	fringe-related protein
At1g05700	leucine-rich repeat protein kinase, putative
At1g09560	GLP5 (GERMIN-LIKE PROTEIN 5); manganese ion binding
At1g09790	COBL6, phytochelatin synthetase-related
At1g18910	zinc finger (C3HC4-type RING finger) family protein
At1g21140	nodulin, putative
<u>At1g23020</u>	ATFRO3 (FERRIC REDUCTION OXIDASE 3); ferric-chelate reductase
At1g34760	GRF11 (General regulatory factor 11), amino acid binding
At1g47400	unknown protein
At1g49000	unknown protein
At1g49820	ATMTK, S-methyl-5-thioribose kinase
At1g56160	AtMYB72 (myb domain protein 72); DNA binding/transcription factor
At1g56430	nicotianamine synthase, putative
At1g60610	protein binding/zinc ion binding, unknown protein
At1g68650	unknown protein
At1g73120	unknown protein
At1g74770	unknown protein
At1g77280	protein kinase family protein
At2g01880	ATPAP7 (purple acid phosphatase 7); serine/threonine phosphatase
At2g02310	ATPP2-B6 (Phloem protein 2-B6)
At2g05830	GTP binding/eukaryotic translation initiation factor/eIF-2B family protein
At2g15830	unknown protein
At2g19410	protein kinase family protein
At2g20030	zinc finger (C3HC4-type RING finger) family protein
At2g28160	ATBHLH029/FIT1/FRU (Fe DEFICIENCY-INDUCED TF 1)
At2g29750	UGT71C1, UDP-glucuronosyl/UDP-glucosyl transferase family protein
At2g30660	3-hydroxyisobutyryl-coenzyme A hydrolase/CoA-thioester hydrolase
At2g32270	ZIP3 (ZINC TRANSPORTER 3 PRECURSOR); zinc ion transporter
At2g35930	PUB, U-box domain-containing protein
At2g36885	unknown protein
At2g40300	ATFER4 (FERRITIN 4); ferric iron binding
At2g45400	BEN1, dihydroflavonol 4-reductase family/dihydrokaempferol 4-reductase
At3g06890	unknown protein
At3g07720	kelch repeat-containing protein
At3g08040	FRD3 (FERRIC REDUCTASE DEFECTIVE 3); antiporter
At3g11750	dihydroneopterin aldolase, putative
At3g12820	AtMYB10 (myb domain protein 10); DNA binding/transcription factor
At3g12900	oxidoreductase, 2OG-Fe(II) oxygenase family protein
At3g13610	oxidoreductase, 2OG-Fe(II) oxygenase family protein
At3g18290	EMB2454 (EMBRYO DEFECTIVE 2454); protein binding/zinc ion binding
At3g18560	unknown protein
At3g21240	4CL2 (4-coumarate:CoA ligase 2); 4-coumarate-CoA ligase
At3g25190	nodulin, putative
At3g46900	COPT2 (Copper transporter 2); copper ion transporter
At3g47640	basic helix-loop-helix (bHLH) family protein
At3g48450	nitrate-responsive NOI protein, putative
At3g50740	UGT72E1 (UDP-glucosyl transferase 72E1)
At3g51200	auxin-responsive family protein
At3g53280	CYP71B5 (CYTOCHROME P450 71B5); oxygen binding
At3g53480	ATPDR9 (PLEIOTROPIC DRUG RESISTANCE 9); ATPase
At3g56090	ATFER3 (FERRITIN 3); ferric iron binding
<u>At3g56980</u>	ORG3 (OBP3-responsive gene 3); DNA binding/transcription factor
At3g58060	cation efflux family protein/metal tolerance protein, putative (MTPc3)
At3g58810	MTPA2; efflux permease/zinc ion transporter
At3g60330	AHA7 (<i>ARABIDOPSIS</i> H(+)-ATPASE 7)
At3g61410	unknown protein
At3g61930	unknown protein
At4g02330	ATPMEPCRB,pectinesterase family protein

(continued on next page)

Table 1 (continued)

ATG	Description
At4g09110	zinc finger (C3HC4-type RING finger) family protein
At4g10510	subtilase family protein
At4g12910	SCPL20 (serine carboxypeptidase-like 20); serine carboxypeptidase
At4g14680	APS3 (ATP sulfurylase 2); sulfate adenyltransferase (ATP)
<u>At4g16370</u>	ATOPT3 (OLIGOPEPTIDE TRANSPORTER); oligopeptide transporter
At4g19370	unknown protein
At4g19680	IRT2 (iron-responsive transporter 2); iron ion transporter/zinc ion transporter
At4g19690	IRT1 (Fe-REGULATED TRANSPORTER 1); Cd, Fe, Mn, & Zn transporter
<u>At4g26890</u>	MAPKKK16 (Mitogen-activated protein kinase kinase kinase 16)
At4g29220	phosphofructokinase family protein
At4g30120	HMA3 (Heavy metal ATPase 3); coupled to transmembrane movement of ions
At4g31940	CYP82C4 (cytochrome P450, family 82, subfamily C, polypeptide 4)
At4g31950	CYP82C3 (cytochrome P450, family 82, subfamily C, polypeptide 3)
At4g33020	ZIP9 (ZINC TRANSPORTER 9 PRECURSOR); cation transporter
At4g38950	kinesin motor family protein, unknown protein
At5g01060	protein kinase family protein
At5g01600	ATFER1 (ferretin 1); ferric iron binding
At5g02780	ln2-1 protein, putative, unknown protein
At5g03570	ATIREG2, iron-responsive transporter-related
<u>At5g04150</u>	BHLH101, basic helix-loop-helix (bHLH) family protein
At5g04730	unknown protein
At5g04950	NAS1, nicotianamine synthase
At5g05250	unknown protein
At5g06490	zinc finger (C3HC4-type RING finger) family protein
<u>At5g13740</u>	ZIF1 (ZINC INDUCED FACILITATOR 1); sugar porter
At5g35580	kinase
At5g36890	glycosyl hydrolase family 1 protein, hydrolyzing O-glycosyl compounds
At5g38820	amino acid transporter family protein
At5g47870	unknown protein
At5g47910	RBOHD (RESPIRATORY BURST OXIDASE PROTEIN D)
At5g53450	ORG1 (OBP3-RESPONSIVE GENE 1); kinase
At5g55620	unknown protein
At5g67330	ATNRAMP4 (natural resistance-associated macrophage protein 4)
<u>At5g67370</u>	unknown protein

regulatory mechanisms and present a more personal view of understudied aspects that need to be approached in the course of reaching this goal.

2. The robust ferrome

The ability of plants to adapt to a continually changing requirement for Fe and to its availability in the soil is fundamental to survival. The earliest adaptive responses to Fe supply have been detected as changes in transcript abundance. Of the several investigations into Fe deficiency-induced changes in the transcriptome, data from five recent reports have been taken to construct a set of transcripts that robustly reflect the transcriptional response of *Arabidopsis* to Fe deficiency [6–10]. A thorough time-course study of transcriptional changes in roots and leaves upon Fe deficiency was conducted by [11], but at this time no whole genome chips were available and this study was not further considered for deciphering the ferrome. After manual removal of transcripts that were annotated to more than one gene, 92 transcripts that are found to be differentially expressed in three of these five array experiments are defined here as the *Arabidopsis* ferrome (Table 1). Consistent with reports found in the literature, the gene ontology terms (GO)

linked to the *Arabidopsis* ferrome are greatly enriched in terms of ion transport, particularly heavy metal cations, and homeostasis of metal cations (Fig. 1). Noteworthy is the fact that many of the transcripts in these categories are not solely associated with Fe but also with Zn, Mn and Cu transport and homeostasis. This indicates that response to Fe deficiency includes a general re-distribution of nutrient cations. Increased metal cation concentrations in shoots and roots have been frequently observed to be correlated with Fe deficiency [12]. These increases may be the result of the lack of specificity of the AtIRT1 transporter in the absence of Fe, as has been suggested [13,14,6 and references therein] or may also be an adjustment in metal cation homeostasis in anticipation of Fe deficiency [10]. The genes included in these over-represented categories belong to the basic response of *Arabidopsis* to Fe deficiency, since they are found to be over-represented in all array experiments regardless of growth conditions (hydroponic or on agar plates) and whether rapid or long-term responses to deficiency were investigated.

Of the 92 genes in the robust ferrome, 57 (62%) have been shown to be differentially regulated under Fe deficiency between the *fit1-1* mutant and the wild-type (Fig. 2, insert). A further subset of genes (15%) have been shown to be differentially regulated under Fe deficiency in the *pye1* mutant compared to the wild-type. There is no overlap between genes regulated by PYE and FIT (Fig. 2, insert). The remaining 21 genes (23%) belonging to the ferrome cannot be assigned to a transcriptional regulon.

The intracellular localization of the proteins encoded by the ferrome is illustrated in Fig. 3. Location was assigned based on experimental evidence found in the literature or on strong predictions

based on protein primary structure (<http://aramemnon.botanik.uni-koeln.de/>). Even using this admittedly liberal criterion, nearly 38% of the ferrome cannot be assigned to a cellular compartment. The most abundant category that can be assigned to a cellular location contains those proteins found in secretory pathways (17% of the ferrome), which is itself a loosely defined compartment including the organelles endoplasmic reticulum, Golgi apparatus and trans-Golgi network. The function of nearly all of the proteins in this category remains to be characterized. Proteins assigned to the vacuolar and plasma membrane compartments (together 21% of the ferrome) have in several cases been investigated in detail. Approximately 14% of the ferrome can be assigned to plastids. This result is perhaps somewhat surprising since the ferrome was determined from root expression data. However, the number of plastid proteins in the ferrome clearly indicates an under-estimated importance of this compartment in the Fe deficiency response. The high percentage of plastid targeted proteins in the root ferrome makes it tempting to speculate that the abundance of this subset of genes is even higher in the less well explored leaf ferrome, possibly representing a conserved response to Fe deficiency.

3. Gene co-expression networks

Most biological functions or regulatory processes cannot be attributed to a single gene. A challenge in deciphering biological processes is the identification of functional modules that orchestrate the processes under investigation. The rapidly growing collection of gene expression profiles allows identification of genes that are closely co-regulated among different conditions and that

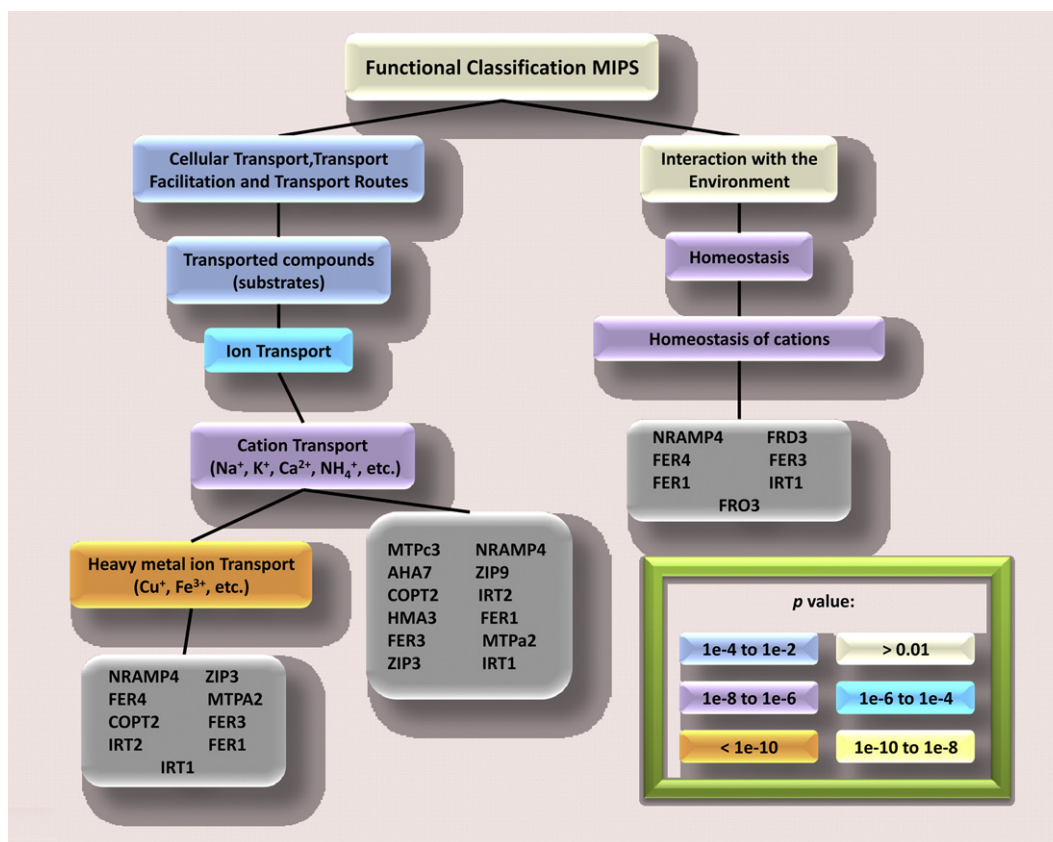


Fig. 1. Gene ontology analysis of the ferrome. Genes belonging to the ferrome were analyzed using the BioMap tool package provided in <http://virtualplant.bio.nyu.edu/>. Functional classification by the Munich Information Center for Protein Sequences (MIPS) was used and *p*-values for over-representation were calculated by the hypergeometric distribution method. Categories with *p* < 0.01 were included in the figure.

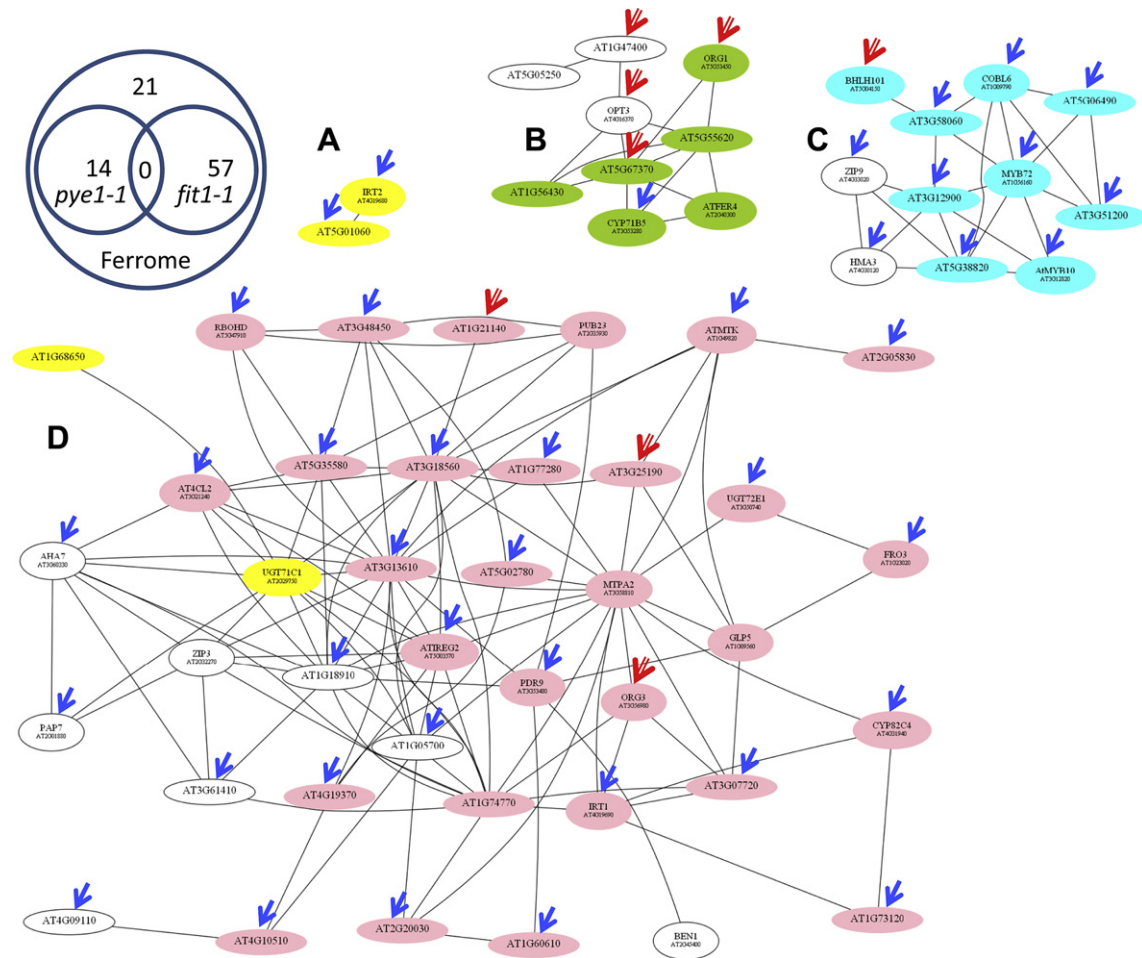


Fig. 2. A co-expression network of the robust ferrome. Co-expression was performed with the MACCU toolbox with a Pearson's correlation of $r > 0.6$. Four clusters were identified (A–D) and nodes were marked in different colors. White nodes indicate genes with relationships that are not root-specific. The cutoff value for unspecific co-expression relationships was $r > 0.5$. Genes in the co-expression network that have been shown to be regulated by FIT are marked with blue arrows and those regulated by PYE are marked with red arrowheads. Insert: The 92-gene ferrome shown as a Venn diagram. Regulation by FIT and PYE was inferred from [7], [12] and [52].

are by inference likely to be functionally related. Networks of co-expressed genes can be generated by calculating the pair wise correlation between genes. Subsequently, networks representing potential interactions are constructed in graphic form in which genes are represented by nodes and interactions between them by edges (Fig. 2). Starting from a population of genes with high stringency, co-expression-based clustering can substantially reduce biological complexity, eliminate noise, and narrow potentially functional linkages between genes. It is important to mention, however, that stochastic fluctuations (i.e. 'noise') may be of crucial importance in biological systems, for example to set up regulatory circuits [15]. The application of this analytical tool is illustrated by a co-expression analysis of the 92 Fe-responsive genes belonging to the ferrome. This analysis identified 61 co-regulated genes that were grouped into four clusters, consisting of between two (Fig. 2A) and 39 (Fig. 2D) genes. The clustering in this analysis was performed with a cutoff value of $r > 0.6$ for the relationship between two genes. The gene clustering was performed with the Multi-Array Correlation Computation Utility (MACCU) tool kit that computes co-expression clusters among any sets of up to several thousands genes using a PC [10]. The software and a thorough description are available at <http://maccu.sourceforge.net/>. Tissue- or organ-specific clustering of microarrays or RNAseq data can further help to remove relationships between genes that are not

relevant for the cell type under investigation and thus helps to reduce biological complexity. On the other hand, a crucial component of a signaling or metabolic pathway might not necessarily show responsiveness to the experimental condition, but rather interact with a partner whose message level is subject to significant changes. Another experimental approach using co-expression relationships is to use one or a subset of genes that are differentially expressed as bait to fish for closely related genes that are not responsive to the experimental conditions. With this approach otherwise overlooked genes may be set in a new context that makes biological sense and can be addressed experimentally.

In the co-expression analysis of the ferrome, the largest cluster containing 39 members, in which FIT-regulated genes were weakly over-represented and PYE-regulated genes were weakly under-represented (Fig. 2D). This central cluster contains only few well-characterized transcripts (e.g. IRT1), with the majority of transcripts being poorly or uncharacterized. The co-expression analysis is useful in identification of transcripts that are basic to the Fe deficiency response. As an illustration of this point, the two transcripts, At1g21140 and At3g25190 (Fig. 2D), have not previously been directly discussed in the deficiency response of *Arabidopsis*. However, recently they have been proposed to be involved in Fe transport into the vacuole [16]. As a second example, the Respiratory Burst Oxidase Protein D (RBOHD, At5g47910) is a protein

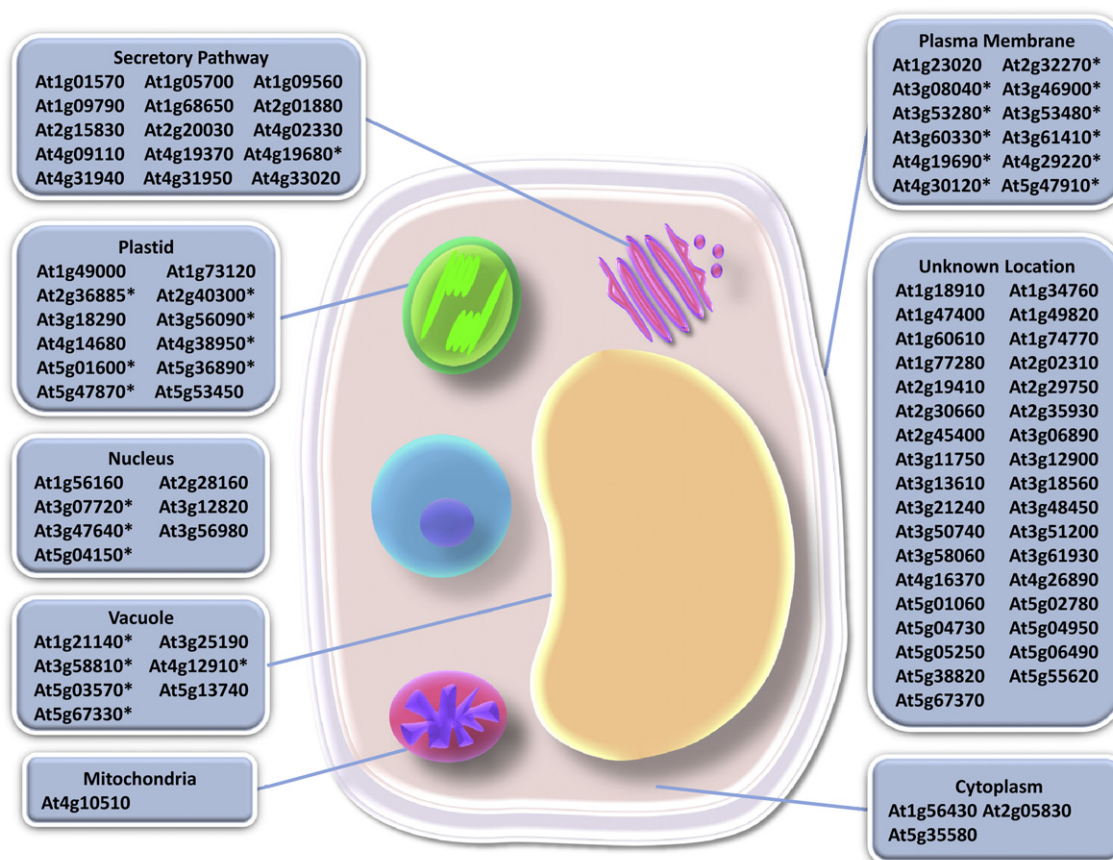


Fig. 3. Intracellular localization of proteins encoded by the 92-gene ferrome. Localization to cellular organelles was assigned to proteins based on experimental evidence reported in the literature or as predicted by <http://aramemnon.botanik.uni-koeln.de/>. In the case of predicted localization, a protein was assigned to a specific organelle only when high scores were obtained from multiple predictors. *Designate experimentally verified locations.

belonging to a family of oxidases that are often associated with pathogen defense. Our co-expression analysis indicates that the function of this member might be more closely related to the response to Fe supply and that a closer investigation of this gene is warranted.

4. More networks: the protein–protein interactome

Similar to co-expression-based gene clustering, networks of protein–protein interactions (PPIs) can be inferred based on predicted or experimental verified interactions (interologs) between proteins. Numerous examples support the observation that protein–protein networks form the basis of networks for most vital cellular processes, including metabolic pathways, signaling cascades and other cellular function. In *Arabidopsis*, an interacting protein was reported to have on average 11 interacting partners [17]. Several studies providing data on protein profile changes upon Fe deficiency are available for tomatoe [18,19], setting the stage for PPI studies in other species. The most conserved interactions were found between proteins with functions in protein turnover and nucleic-acid-related processes. Interestingly, highly connected hubs appear to be more conserved than hubs with a few interacting partners, suggesting a higher evolutionary pressure on the former group. Protein–protein interactions can be a valuable source of information to assign functions for unknown proteins based on their interacting partners and to add new components to signaling networks or metabolic pathways. Although a significant trend of interacting proteins being co-expressed was reported [17], the

multitude of post-translational protein modifications makes it tempting to assume that in many cases at least one partner of an interolog pair may not be differentially expressed at the transcriptional level. Protein–protein interactions networks may also provide an indispensable tool to uncover new layers of regulation in cellular Fe homeostasis. An easy-to use tool for mining of PPIs is the interactome 2.0 database (http://bar.utoronto.ca/interactions/cgi-bin/arabidopsis_interactions_viewer.cgi).

5. Epigenetic regulation

In eukaryotic cells, the higher order of chromatin structure is critical for the efficiency of gene expression. The reversible acetylation of histones is catalyzed by histone acetyl transferases and is strongly correlated with gene expression. The reverse process is catalyzed by histone deacetylases, often acting in concert with histone methyltransferases and resulting in transcriptional repression (Fig. 4). The effects of histone modifications can be twofold: firstly, transcription can be affected by structural alterations, and secondly histone-related processes can facilitate the recruitment of repressors or activators of gene activity. Chromatin modifications (CM) such as DNA methylation have been implicated in abiotic stress responses [20], but little is known about the role of CM in nutrient signaling. In yeast, histone acetylation via Gcn5 and the SWI/SNF nucleosome remodeling complex were shown to be required for several genes related to phosphate uptake [21–24]. A chromatin-level control has been recently reported for the phosphate deficiency response of *Arabidopsis* [25]. Proper

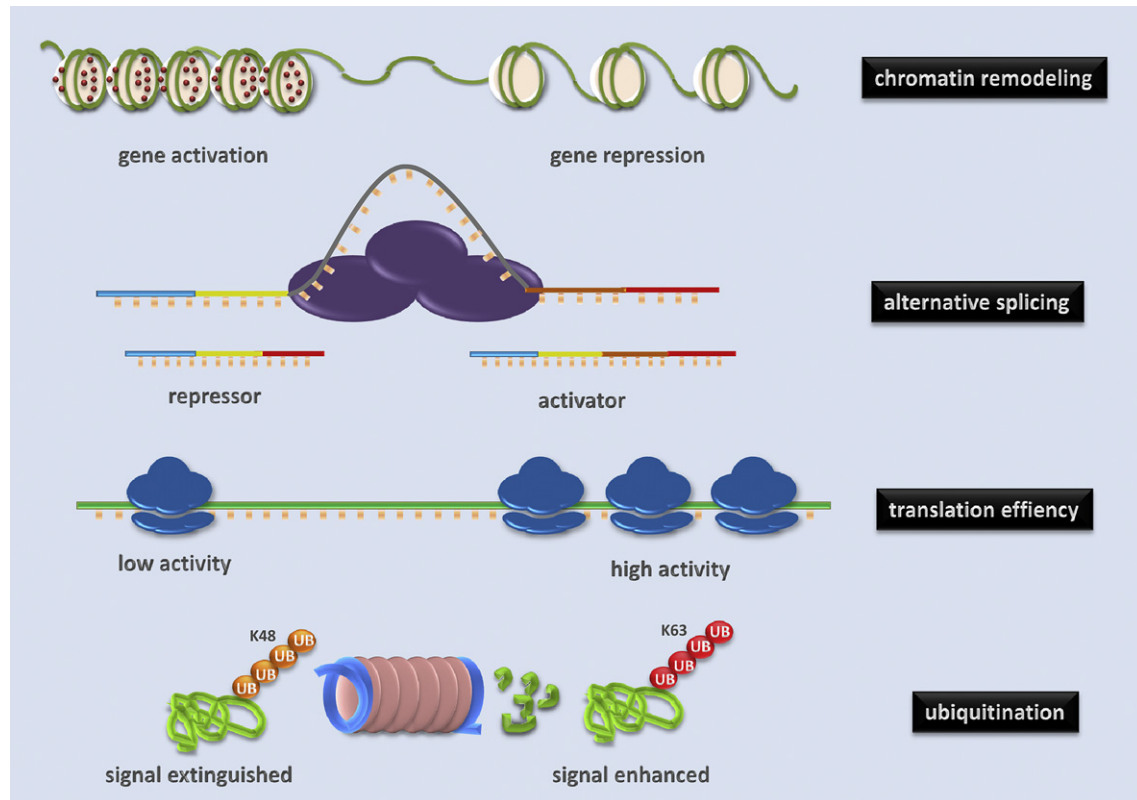


Fig. 4. Possible levels of regulatory control. Chromatin remodeling can regulate gene expression by histone methylation (symbolized by red dots) in concert with histone deacetylation. Acetylation of histones increases accessibility of the DNA and is associated with gene activation. Alternative splicing can produce different transcript isoforms from the same gene, possibly resulting in dysfunctional proteins or proteins with altered activity. Association of RNA with polysomes can selectively support the translation of a subset of transcripts. Polyubiquitin chains linked via Lys 48 targets proteins for degradation via the 26S proteasome. Lys 63 polyubiquitin chains may alter the activity of proteins independent from proteolysis and may be involved in signal transduction.

deposition of the histone variant H2A.Z by the SWR1 chromatin remodeling complex at target loci is required for control of a suite of Pi starvation response genes. Loss of H2A.Z is associated with the manifestation of multiple phosphate starvation responses, indicating a crucial function of the SWR1 complex in phosphate homeostasis in *Arabidopsis*. No information about a putative role of chromatin modifiers for cellular Fe homeostasis is available. It is interesting to note, however, that chromatin methylation/demethylation is dependent on Fe-containing enzymes [26], making a putative link between Fe status and chromatin remodeling.

6. Post-transcriptional regulation: the intervening layer

6.1. RNA processing I: lost in translation?

Following transcription, RNA undergoes several modifications, including addition of a 5' cap, a 3' polyA tail, and removal of introns by the spliceosome. While transcriptional regulation is well explored, the molecular mechanisms of post-transcriptional control are less well understood. Transcriptional profiling approaches do not distinguish between mRNA populations that undergo active transcription (i.e. associated with polyribosomes) and other messenger ribonucleoproteins (mRNPs) that mediate storage or degradation. As a fast response to external or internal signals, translation of a subset of mRNAs can be reduced or inhibited by the formation of stable mRNPs that prevent both translation and degradation. These mRNAs could subsequently associate with ribosomes and be translated. In this way, translation is effectively regulated without the necessity of a cycle of degradation and *de novo* synthesis. Messenger ribonucleoprotein

homeostasis is highly dynamic in metabolically active cells, and the composition of mRNPs and thus the fate of mRNAs can rapidly change [27]. Ribosome loading of individual mRNAs can be re-prioritized in response to environmental conditions and may significantly alter the protein profile of organs or tissues. Environmental conditions affect the translation of individual mRNAs [28–30], fine-tuning translation, which is independent of the transcriptional activity. For example, transcript abundance may be altered without affecting the amount of mRNA of the respective gene being associated with polysomes, or a gene may be differentially accumulated in polysomal RNA with or without changes in total RNA.

Large-scale analysis of mRNA translation states has been conducted during dehydration [31], sucrose starvation [29], and oxygen deprivation [32]. In all cases, most of the differentially expressed genes showed a decrease in polysomal RNA. While the translome (i.e. the profile of ribosome-associated mRNAs) of Fe-deficient *Arabidopsis* roots has not yet been investigated, it is tempting to assume that this may also be the case for Fe-limited plants.

A further potential regulatory switch is represented by the composition of ribosomes. Cytosolic ribosomes are comprised of two subunits composed of 81 ribosomal proteins (RPs) encoded by 251 genes [33]. RPs are classified as housekeeping, but this view was recently challenged. Several RP mutants are affected in cell division and/or cell expansion resulting in deformed leaves, indicating specific function of some RPs in developmental processes. Furthermore, similar to the 26S proteasome (see below), current thinking suggests that the composition of RP is highly dynamic and is possibly affected by environmental signals [34]. It is thus tempting to speculate that differential expression of RP can control

mRNA preference and thus contribute to translation efficiency. Similar to other RNA-related processes, this would allow fast regulation without the necessity of going through a cycle of degradation and *de novo* synthesis.

6.2. RNA processing II: alternative splicing

Alternative splicing (AS) generates more than one mature mRNA from a single gene by differentially splicing of introns and exons. In eukaryotes, AS is a major route of post-transcriptional regulation, contributing to protein diversity by yielding proteins that differ in function, activity, binding properties, sub-cellular localization or other properties. In addition, AS can produce mRNAs that contain premature termination codons and are subject to nonsense-mediated decay. The major types of AS are intron retention, alternative donor/acceptor, and exon skipping (see [35] for a comprehensive review). In humans, the majority of genes undergo AS with estimates of up to 88% of the total genes. In plants (*Arabidopsis* and rice), the estimated number has been corrected several times, ranging from 22% [36] up to 42% of intron-containing genes in a recent study in *Arabidopsis* using RNAseq [37]. The splicing of introns is mediated by the spliceosome, a complex macromolecular machine [38]. AS has not been extensively studied in plants. In contrast to humans, in which the most frequent form of AS is exon skipping, the most common form of AS in plants is intron retention. Due to their sessile life form, the response of plants to the environment is more complex than in animals, and an increase in the number of alternatively spliced genes upon exposure of the plants to a certain condition may be anticipated from the growing pool of data from tiling array and RNAseq studies. Cross-species analysis of alternative spliced orthologs revealed a lack of conservation of AS-induced polymorphisms, indicating a limited role of AS in functional expansion of the protein [39]. It should be considered, however, that AS may be restricted to a developmental stage or associated with a particular tissue, rendering a cross-species comparison difficult.

Whether induced AS represents an acclimation to environmental conditions or rather an increased RNA processing error rate due to inaccurate recognition of splicing sites remains to be elucidated. Numerous studies have shown that AS can be influenced by abiotic factors [40]. Potentially, AS could generate many new proteins some of which might acquire a new function and could be of immediate advantage in acclimation to new environmental conditions. In a recent analysis of Fe-deficient *Arabidopsis* roots using RNAseq, we found some ca. 250 transcripts to contain differentially induced alternate exons (i.e. the transcripts were not alternatively spliced under control conditions; Ray, Lin and Schmidt, unpublished). A subset of these genes was also found to be differentially alternatively spliced either upon Pi or Mn starvation, indicating that at least some, if not the majority of these transcript are due to authentic AS events and not to noise related to mRNA processing. Only a very small subset of the AS genes were also differentially expressed upon Fe deficiency. It is thus tempting to speculate that AS represents an unexplored level of regulation of the strategy I type Fe deficiency response, adding new components to the puzzle of known players. A future challenge is the functional validation of predicted Fe deficiency-induced AS. An even bigger challenge is to understand the regulation of AS and to decipher how AS is integrated into other regulatory mechanisms such as chromatin modification and transcription.

7. Post-translational modifications of proteins - canonical versus non-canonical ubiquitination

In a wide variety of pathways and signal transduction chains, the fate of proteins is controlled via modifications by ubiquitin.

Ubiquitin is first activated by an ubiquitin-activating enzyme (E1), transferred to a ubiquitin-conjugating enzyme (E2), and subsequently conjugated to the substrate protein by a ubiquitin-protein ligase (E3). Proteins labeled with polyubiquitin chains linked through K48 are recognized by regulatory subunits of the 26S proteasome and targeted for degradation [41]. Polyubiquitin chains linked to substrates through K63 regulate protein activity in a non-proteolytic manner. Both degradative K48-linked and non-canonical K63-linked polyubiquitin chains have an essential role in cellular signaling. In mammalian cells, cellular Fe homeostasis is regulated by the RNA-binding proteins Fe regulatory protein 1 and 2 (IRP1 and IRP2), which control the translation and stability of mRNAs encoding proteins that are required for Fe uptake and storage. IRP1 is a bifunctional protein that under Fe-replete conditions, functions as a cytosolic aconitase. Under Fe-limited conditions, an 4Fe-4S cluster that is essential for the aconitase function disassembles and the protein acquires IRP activity. A proteolytic pathway links the degradation of IRP2 by the SKP1-CUL1-FBXL5 protein complex to the intracellular Fe concentration [42], representing a key regulatory step in cellular homeostasis in animal cells. In plants, a regulation of Fe homeostasis via IRPs is not operational [43], an involvement of ubiquitin-dependent proteolysis may, however, be safely assumed. In *Arabidopsis*, overexpression of the two major proteins involved in Fe acquisition, the ferric oxidoreductase FRO2 and the Fe transporter IRT1 results in an increase in message level, but not to an increase in protein or activity level [44,45]. It is tempting to speculate that ubiquitin-dependent proteolysis provides a fast means for the plants to avoid the accumulation of toxic amount of Fe upon sudden changes in availability. BRUTUS (BTS), a putative E3 ligase, was shown to be co-regulated with POPEYE, a newly identified transcription factor which plays a critical role in Fe homeostasis [14]. Roots of *bts* mutant plants were found to have a higher acidification response when compared to the wild-type and were less susceptible to Fe starvation, indicating that BTS is a negative regulator of the Fe stress response. While the mode of BTS action has not yet been elucidated, ubiquitin-dependent degradation of proteins positively regulating Fe acquisition seems to be a plausible scenario.

In a proteomics approach, we recently identified an E2 conjugase, *UBC13*, that catalyzes K63-linked polyubiquitylation in cucumber root tips [46]. Lysine 63 chain-forming enzymes have been identified in several regulatory contexts in animals and yeast, including signaling from membrane receptors. Ectopic expression of the cucumber *UBC13* gene in *Arabidopsis* led to a more pronounced formation of branched root hairs upon Fe starvation, a key response of *Arabidopsis* roots to Fe deficiency. Plants carrying a mutation in the *Arabidopsis* ortholog *UBC13A* were unable to form branched root hairs upon Fe deficiency and showed a perturbed expression of Fe-regulated genes. Mutants defective in both *Arabidopsis* *UBC13* genes, *UBC13A* and *UBC13B*, showed a marked reduction in root hair density. Defects in the cognate E3 ligases *RGLG2* and *RGLG1* caused the constitutive formation of branched root hairs independent of the Fe supply, indicating the involvement of polyubiquitination in the altered differentiation of rhizodermal cells.

The 26S proteasome is composed of a 20S core protease (CP) and two 19S regulatory particles that recognize and transport substrates into the CP for degradation. Recent research suggests that the 26S proteasome is a highly dynamic structure, assembled with paralogous pairs for most units which are subject to post-translational modifications such as ubiquitination and or acetylation [47]. Some subunits of the regulatory particle have been shown to play crucial roles in growth and development and may be involved in the regulation of stress responses [48,49]. The non-ATPase regulatory particle subunit RPN12a is targeted by the

U-box-containing E3 protein ligases PUB22 and PUB23, resulting in decreased ubiquitin-dependent proteolysis [50]. PUB23 is robustly up-regulated by Fe deficiency (Table 1; [10]), making it tempting to suggest that ubiquitin-dependent proteolysis can be regulated by ubiquitination of the regulatory particle. It has been speculated that RPN12a is polyubiquitinated by K63-linked ubiquitin chains [50], which would provide a link between UBC13, PUB23 and the regulation of a subset of Fe-responsive genes.

8. Conclusions

While it took ‘Deep Thought’ seven and a half million years to come up with the answer to ‘life, the universe and everything’ in Douglas Adams’s novel, it took less than a decade to identify and decipher precise functions of several genes of the ferrome. The categorization of proteins, metabolites and transcript that are altered in abundance upon Fe deficiency, however, marks only the starting point to understand the complex response of plants to Fe deficiency. The major challenge is to decode the mechanisms and kinetics of regulatory networks that are dictating cellular Fe homeostasis. In addition, intervening layers of regulation deserve our serious attention. These layers appear to be closer to each other than previously thought. For example, splicing occurs co-transcriptionally, and histone modification such as acetylation, methylation, phosphorylation and ubiquitination has been shown to affect exon selection in humans [51]. Many important regulatory players might have been overlooked in transcriptional and proteomics profiling studies. The new tools for RNA sequencing and sensitive techniques for protein profiling with cell-specific resolution are at hand. These tools have already provided valuable information and in the near future will continue to greatly enhance our understanding on how plants adjust Fe uptake to Fe supply. We are confident that the application of this knowledge will facilitate the production of Fe fortified plants and thus improve agricultural productivity and food forage quality.

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